PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	**Believer Agbodeka**
Program/Batch	B084- Networkwalks
Date	12th November 2026
Modules Completed	W2-PM1- (Multiple Kali Tools) W2-pm5 (Zenmap Scanning)
Client/Target	1 Networkwalks (Secured written permission already 2 My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1 Reconnaissance & Footprinting Phase 2 Scanning & Network Discovery

1. Liability Disclaimer

I have performed these activities only on the system & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the Law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loose of your job and a permanent record. In most countries unauthorized access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmpa (w2-pm5). One module covers the footprinting phase and the other covers the scanning phase, so together show how an attacker moves from gathering public information to mapping live host on a network. It is the Week 2 part of my ongoing internship program at Networkwalks. All command were run in Kali Linux (footprinting) and on a windows pc with Zenmap installed (scanning). Every steps below includes the exact command used, the result I observed, a screen short as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and it purpose.

Tool	**PURPOSE**
Kali Linux & Windows	Operating system used for reconnaissance activities
WHOIS	Find domain registration details (owner, date, name servers

Whatweb	Fingerprint web technologies (servers, CMS, plugins, IP)
Nslookup	Resolve the domain name to it IP address using DNS.
Curl	Read the HTTP response headers of the website.
Wafw00f	Detect whether a Web Application Firewall protects the site.
Dnsrecon	Enumerate all the DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs, MAC addresses.
Windows CMP	Local IP and MAC address identification.

4. Activities performed

4.1 Footprintinf & Reconnaissance

I performed reconnaissance against the networkawalks.con domain using six Kali Linux tools: WHOIS, Whatweb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool was used to collect a different type of information about the target.

First, I used WHOIS to obtain publicly available domain registration information and identify the domain's name servers. The results proved information about the domain registration information and hosting infrastructure.

I then used whatweb to identify technologies used by the website. The results identified WordPress 7.0.4 and WP Dowload Manager 3.3.58 along with other information exposed by the website.

Using Nslookup, I resolved the domain name to its IP address. The provided results identified 192.232.216.135

I used Curl with –I option to inspect the HTTP response headers. This proved additional information about the web application and exposed the WordPress Rest API endpoint /wp-json/

Next I used Wafw00f to determine whether a Web Application Firewall was protecting the website. The result identified Mod Security (spiderLabs).

Finally, I used DNSRecon to enumerate DNS records. The results provided information relating to name servers, mail servers, SPF /TXT records service records, and DNS software information.

4.2 Network Scanning With Zenmap

For the second activity, I used Zenmap to preform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the windows command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and select ping scan to identify active hosts.

The example results proved in the practical identified four live hosts:

192.168.172.1

192.168.172.2

192.168.172.128

192.168.172.254

The example results also included four MAC addresses.

After completing the scan, I opened the topology section in Zenmap to enable the legend and saved the network topology in PDF format as required by the practical task.

5. Risk Analysis/ Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risk

#	Risk/Find	Evidences/Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Dowload Manager	Attacker may use exposed technology/version information to identify software requiring further security review	Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Proved information about the network location of the web service	Low
3	HTTP technical information exposed	Cur returned HTTP response headers and exposed WP-json/	May assist technology fingerprinting and further enumeration	Low
4	WAF technology identifiable	Wafw00f identified Modsecurity (spiderLabs)	Reveals information about the web application's security architecture	Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail, and service-related records	DNS information can help build a broader infrastructure profile	medium
6	Multiple live hosts visible on the local network	Zenmap identified four live hosts in the example network	Unknoww or unauthorized devices may potentially be present on a network	Medium

Risk Level Key: .Critical .Medium .Low

The risk above are observation from footprinting and scanning exercise, not confirmed vulnerabilities.

The practical primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of this two modules. Therefore the presence of information as software version, IP address or DNS records does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6 .Recommendations

Based on the observation from these activities, I recommend the following security improvement:

- **Review publicly exposed technology information**
  Organization should regularly review what information about their web technologies, CMS and plugins public visibility.

- Keep software updated
    - CMS platforms, plugins and other web technologies should be regularly updated and review against current security advisories.
- Review HTTP headers
    - HTTP response headers should be reviewed to determine whether unnecessary technical information is be exposed.
- Review DNS records regularly
    - DNS record should be checked periodically to ensure that only required information and services are publicly exposed
- Properly configure and monitor the WAF
    - Keep the WAF (ModSecurity) enable and tuned, since it already blocks naïve attacks.
- Perform regular internal network discovery
    - Organization should periodically scan their own networks to identify active devices.
- Investigate unknown devices
    - Any unexpected devices discovered during the network scanning should be investigated and verified.
- Maintain network documentation
    - Network topology and device information should be documented and updated regularly.
- Perform security testing with authorization
    - Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During week 2 of my cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance, and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can proved domain information, WhatWeb can identify web technologies, Nslookup can resolved domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

In the network scanning activity, I uses Zenmap to identified my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology. The exercise show me that information gathering is an important part of cybersecurity. Even before attempting to exploit

a System, security professional can learn a significant amount about an environment by carefully analyzing

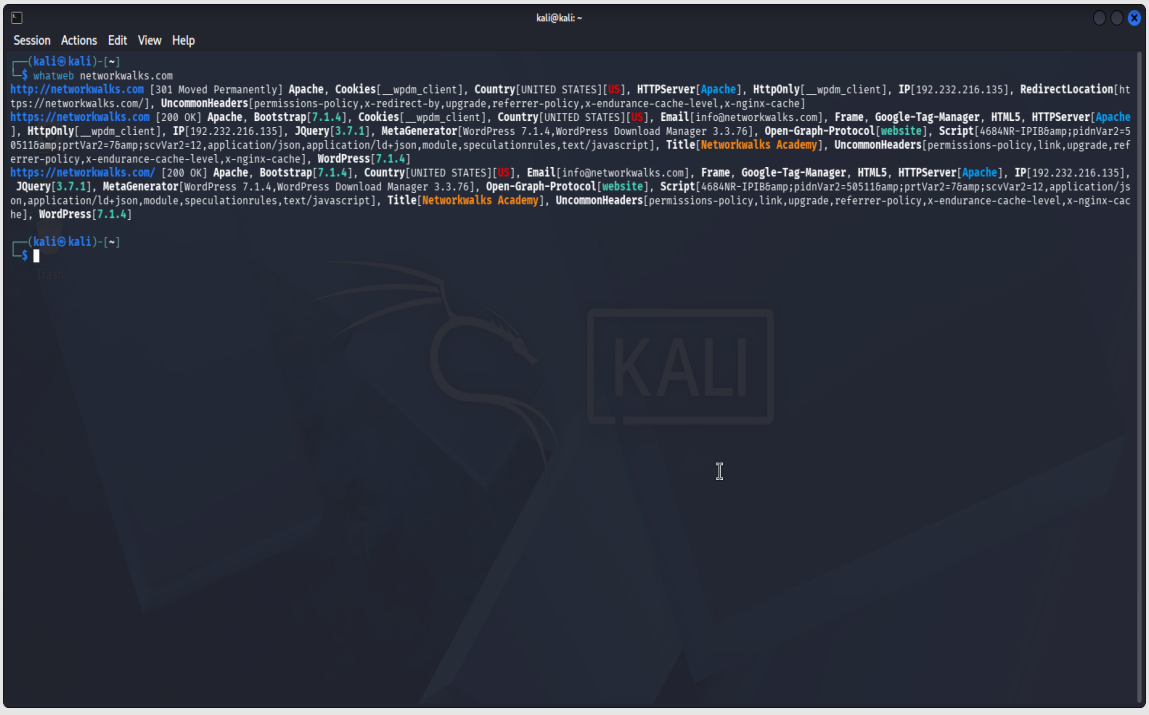
Publicly available information and network responses.

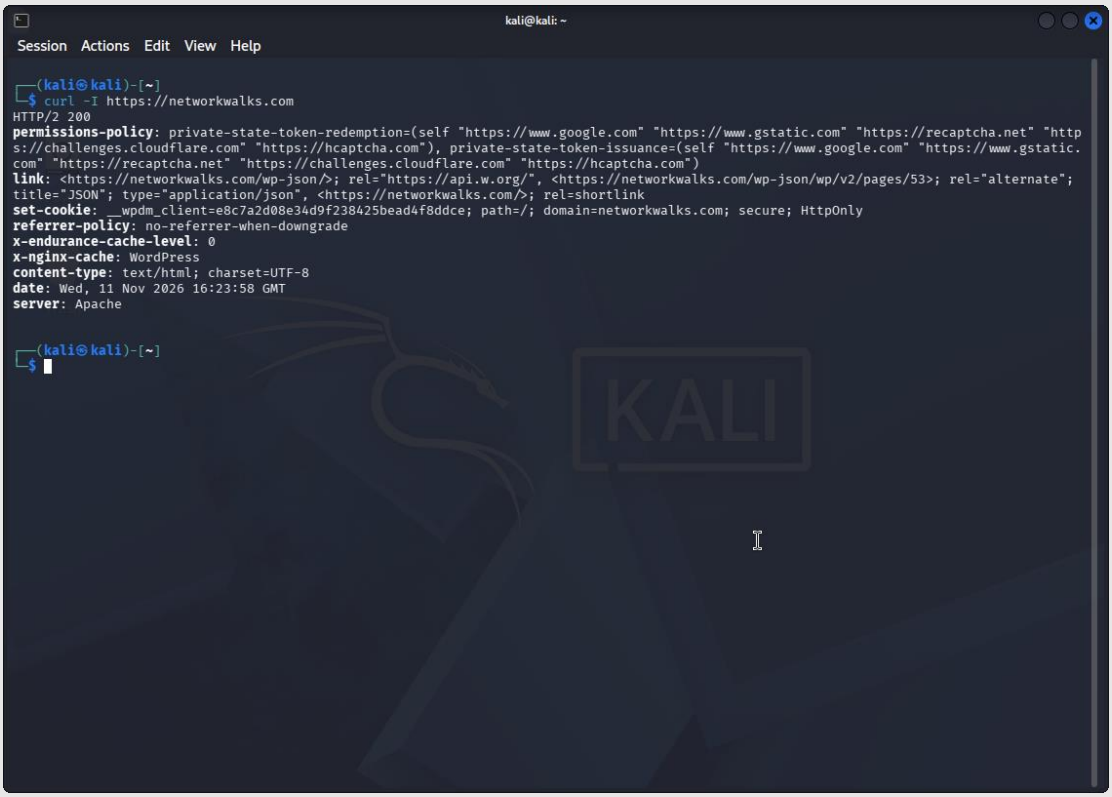
I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

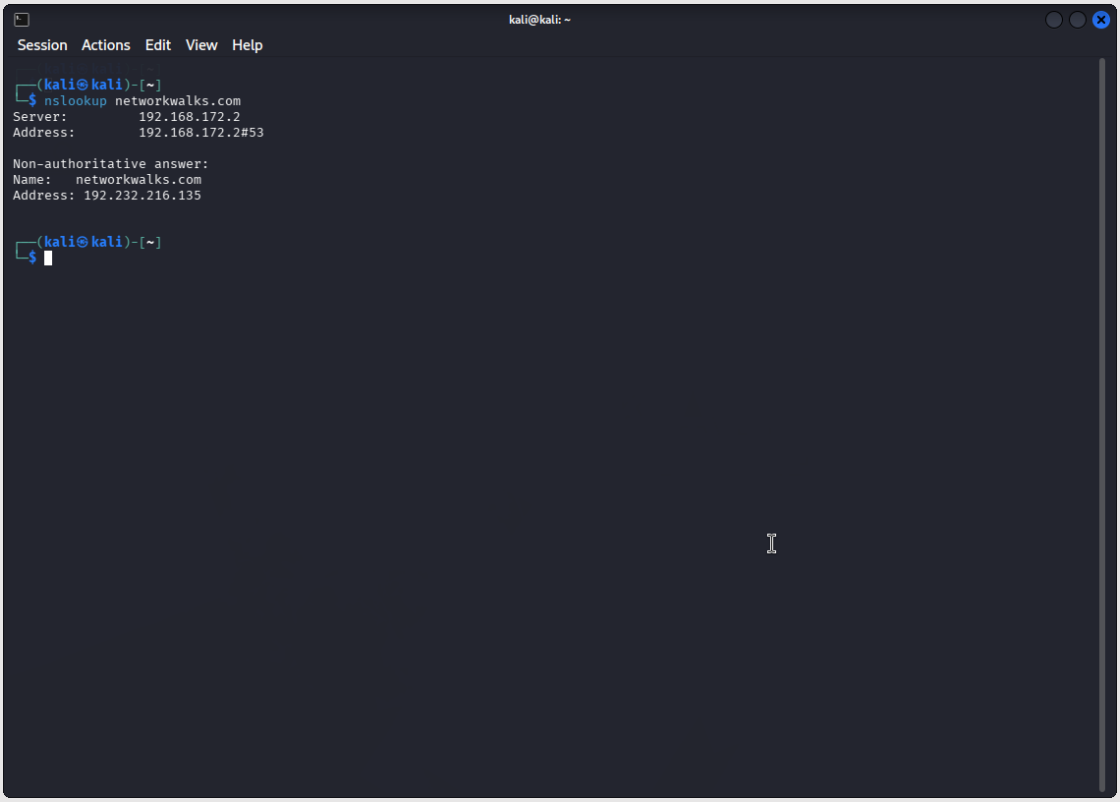
Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity Lab.

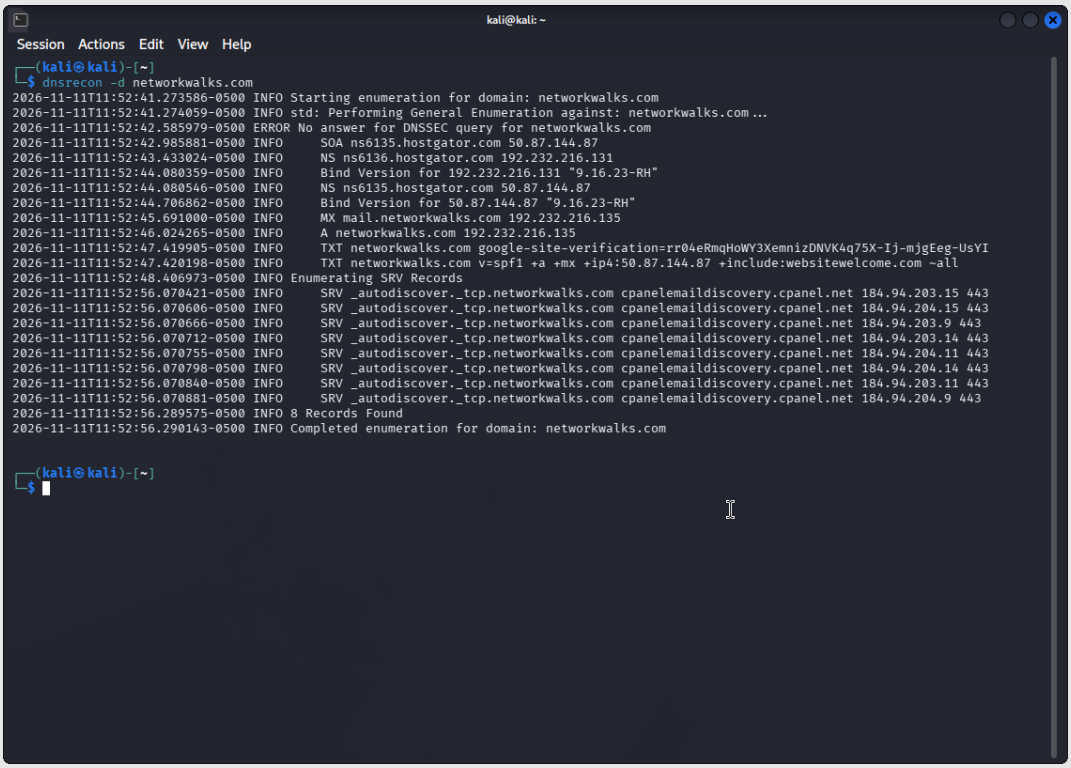
8. Evidences Collected



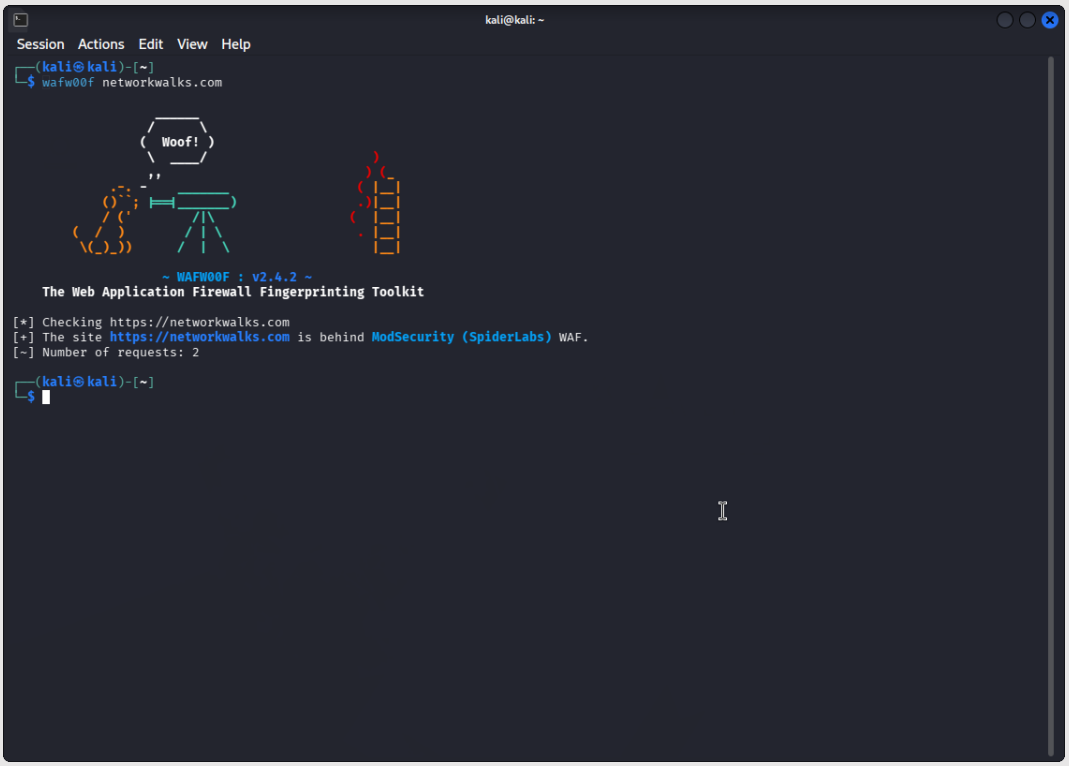


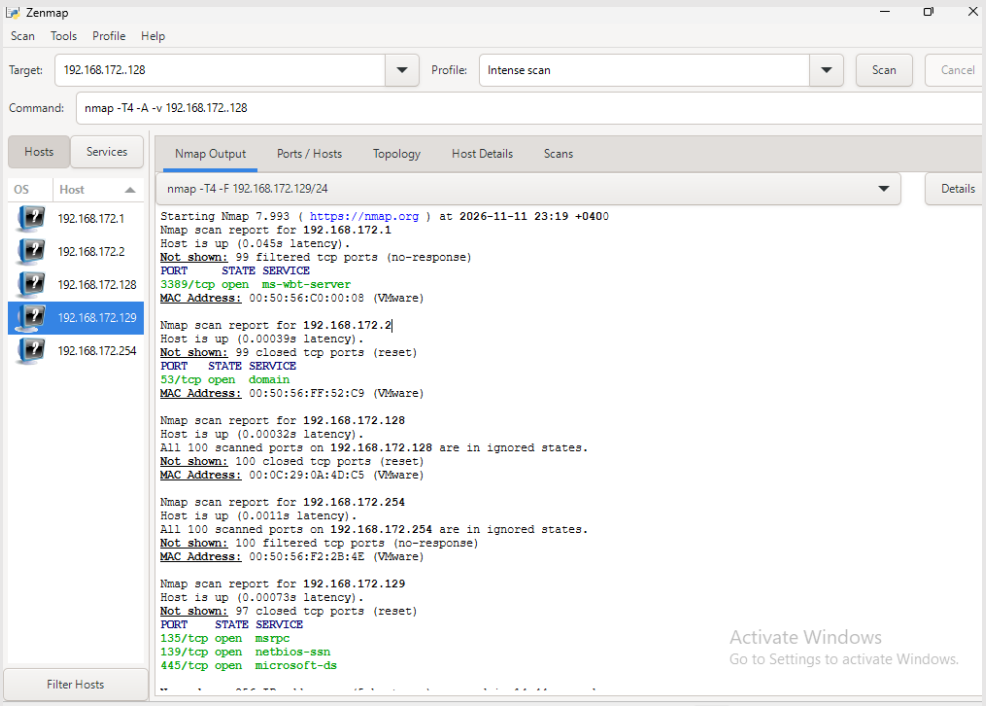

kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=e8c7a2d08e34d9f238425bead4f8ddce; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Wed, 11 Nov 2026 16:23:58 GMT
server: Apache
┌──(kali㉿kali)-[~]
└─$

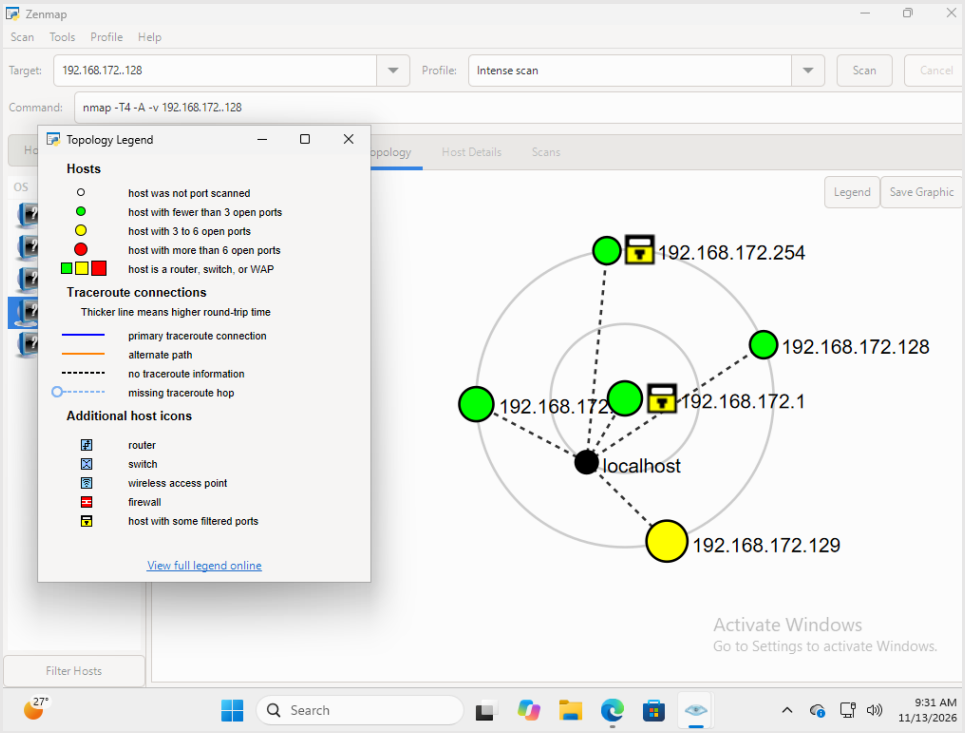

kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ nslookup networkwalks.com
Server: 192.168.172.2
Address: 192.168.172.2#53
Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135
┌──(kali㉿kali)-[~]
└─$


kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ dnsrecon -d networkwalks.com
2026-11-11T11:52:41.273586-0500 INFO Starting enumeration for domain: networkwalks.com
2026-11-11T11:52:41.274059-0500 INFO std: Performing General Enumeration against: networkwalks.com...
2026-11-11T11:52:42.585979-0500 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-11T11:52:42.985881-0500 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-11-11T11:52:43.433024-0500 INFO NS ns6136.hostgator.com 192.232.216.131
2026-11-11T11:52:44.080359-0500 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-11-11T11:52:44.080546-0500 INFO NS ns6135.hostgator.com 50.87.144.87
2026-11-11T11:52:44.706862-0500 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-11-11T11:52:45.691000-0500 INFO MX mail.networkwalks.com 192.232.216.135
2026-11-11T11:52:46.024265-0500 INFO A networkwalks.com 192.232.216.135
2026-11-11T11:52:47.419905-0500 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-11T11:52:47.420198-0500 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-11-11T11:52:48.406973-0500 INFO Enumerating SRV Records
2026-11-11T11:52:56.070421-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.15 443
2026-11-11T11:52:56.070606-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.15 443
2026-11-11T11:52:56.070666-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443
2026-11-11T11:52:56.070712-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.14 443
2026-11-11T11:52:56.070755-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.11 443
2026-11-11T11:52:56.070798-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.14 443
2026-11-11T11:52:56.070840-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
2026-11-11T11:52:56.070881-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.9 443
2026-11-11T11:52:56.289575-0500 INFO 8 Records Found
2026-11-11T11:52:56.290143-0500 INFO Completed enumeration for domain: networkwalks.com
┌──(kali㉿kali)-[~]
└─$

,


kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ wafw00f networkwalks.com
Woof!
~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
┌──(kali㉿kali)-[~]
└─$


Zenmap
Scan Tools Profile Help
Target: 192.168.172..128
Profile: Intense scan
Scan
Cancel
Command: nmap -T4 -A -v 192.168.172..128
Hosts
Services
OS
Host
192.168.172.1
192.168.172.2
192.168.172.128
192.168.172.129
192.168.172.254
Filter Hosts
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -T4 -F 192.168.172.129/24
Details
Starting Nmap 7.993 (https://nmap.org) at 2026-11-11 23:19 +0400
Nmap scan report for 192.168.172.1
Host is up (0.045s latency).
Not shown: 99 filtered tcp ports (no-response)
PORT STATE SERVICE
3389/tcp open ms-wbt-server
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.172.2
Host is up (0.00039s latency).
Not shown: 99 closed tcp ports (reset)
PORT STATE SERVICE
53/tcp open domain
MAC Address: 00:50:56:FF:52:C9 (VMware)
Nmap scan report for 192.168.172.128
Host is up (0.00032s latency).
All 100 scanned ports on 192.168.172.128 are in ignored states.
Not shown: 100 closed tcp ports (reset)
MAC Address: 00:0C:29:0A:4D:C5 (VMware)
Nmap scan report for 192.168.172.254
Host is up (0.0011s latency).
All 100 scanned ports on 192.168.172.254 are in ignored states.
Not shown: 100 filtered tcp ports (no-response)
MAC Address: 00:50:56:F2:2B:4E (VMware)
Nmap scan report for 192.168.172.129
Host is up (0.00073s latency).
Not shown: 97 closed tcp ports (reset)
PORT STATE SERVICE
135/tcp open msrpc
139/tcp open netbios-ssn
445/tcp open microsoft-ds
Activate Windows
Go to Settings to activate Windows.


Zenmap
Scan Tools Profile Help
Target: 192.168.172..128
Profile: Intense scan
Scan
Cancel
Command: nmap -T4 -A -v 192.168.172..128
Topology Legend
Hosts
host was not port scanned
host with fewer than 3 open ports
host with 3 to 6 open ports
host with more than 6 open ports
host is a router, switch, or WAP
Traceroute connections
Thicker line means higher round-trip time
primary traceroute connection
alternate path
no traceroute information
missing traceroute hop
Additional host icons
router
switch
wireless access point
firewall
host with some filtered ports
View full legend online
Legend
Save Graphic
192.168.172.254
192.168.172.128
192.168.172.1
localhost
192.168.172.129
Filter Hosts
Activate Windows
Go to Settings to activate Windows.
Search
9:31 AM
11/13/2026